## Highlights

- Well-to-wake distance-band model links SAF supply risk to ATM and fleet levers
- A one-year SAF delay raises 2050 emissions sharply but cumulative ones only slightly
- ATM measures are the cheapest corrective lever but close only about half the gap
- Long-haul early retirement abates most per aircraft and per euro
- Pathways that pre-commit delivery slots lose corrective headroom

# Absorbing sustainable aviation fuel shortfalls in European aviation: a distance-band model of corrective action through air traffic management, fleet renewal and alternative supply

Fabricio Niebles-Atencio[a,*], Lucía Rivera-Charún[b]

[a]*Institute of Information Systems, Hamburg Business School, University of Hamburg, Von-Melle-Park 5, 20146 Hamburg, Germany*
[b]*Departamento de Ingeniería Civil y Ambiental, Universidad de Ingeniería y Tecnología, Jr. Medrano Silva 165, Barranco, 15063 Lima, Peru*

**Abstract**

The ReFuelEU Aviation regulation obliges fuel suppliers to deliver rising minimum shares of sustainable aviation fuel (SAF), but announced production capacity has repeatedly failed to materialise on time. We ask how much SAF shortfall European decarbonisation pathways can absorb and which corrective action restores them at least cost. A well-to-wake model disaggregated into four flight-distance bands represents air traffic management (ATM) inefficiency, fleet vintage, SAF supply and zero-emission propulsion with band-specific feasibility limits. The base-year flight population is generated by a physically based fuel model calibrated to EUROCONTROL flight and emission shares and checked against independent data. The SAF mandate delivers 92% of the 2050 reduction relative to a no-new-measures reference; zero-emission propulsion contributes 3%. Because the mandate rises in steps, a one-year delay of SAF supply raises 2050 emissions by 52% but cumulative 2025–2050 emissions by only 2.7%, so terminal-year targets are far more fragile than carbon budgets. A linear programme combining ATM measures, band-targeted early aircraft retirement and alternative fuel supply shows that, after an 18% supply shortfall, the cumulative budget can be restored at a net saving of EUR 7.1 bn under the assumed prices, through ATM measures and early retirement of long-haul aircraft; also restoring the 2050 level costs EUR 17.2 bn. With multi-year delays most of the 2050 gap cannot be closed within delivery-slot and supply limits, and a pathway that already

*Corresponding author.
*Email addresses:* fabricio.niebles.atencio@uni-hamburg.de (Fabricio Niebles-Atencio), mriverac@utec.edu.pe (Lucía Rivera-Charún)

commits delivery slots to accelerated renewal leaves 50% of its 2050 gap open even without delay. Implications for SAF target design, ATM regulation and fleet planning follow.

*Keywords:* Sustainable aviation fuel, ReFuelEU Aviation, Air traffic management, Fleet renewal, Linear programming, Decarbonisation pathways

## 1. Introduction

European aviation is expected to reduce its emissions while traffic grows by about 1.6% per year to 2050 (EUROCONTROL, 2024). The sector's abatement levers differ in speed, cost and reach. Sustainable aviation fuels (SAF) can be used by every existing aircraft, which is why the ReFuelEU Aviation regulation places the largest share of the effort on them: fuel suppliers at Union airports must supply minimum shares of 2% in 2025, 6% in 2030, 20% in 2035, 34% in 2040, 42% in 2045 and 70% in 2050, with a synthetic-fuel sub-mandate reaching 35% (European Union, 2023). Fleet renewal lowers fuel burn per kilometre but is bounded by aircraft manufacturing capacity. Operational efficiency in air traffic management (ATM) is fast to deploy but small in scale, and zero-emission propulsion is limited by range to short flights (Akinola et al., 2025; Uddin and Wang, 2025).

Two features of the problem are poorly represented in existing assessments. The first is that the levers act unevenly across flight distances. Flights shorter than 500 km account for about 24% of European departures but only 3.8% of $CO_2$, whereas flights longer than 4000 km account for 6% of departures and 48% of $CO_2$ (EUROCONTROL, 2021). ATM inefficiency is concentrated in the terminal phases and is therefore a larger share of fuel on short flights; zero-emission propulsion is feasible only there; and SAF and fleet renewal are the only levers that reach the long-haul flights that dominate the footprint. A model that treats the system as homogeneous cannot represent these interactions.

The second feature is supply risk. Of the SAF capacity announced for 2024 only about a quarter was operational on time, and modelled EU shortfalls range from 18–43% of the 2030 mandate and 5–35% of the 2050 mandate (Martulli et al., 2025). A mandate expressed as a minimum share is binding on paper but bounded in practice by the fuel that is available. Robust decarbonisation strategies under background uncertainty have been studied at

the system level (Costa-Alves et al., 2026; Dray et al., 2022; Bergero et al., 2023), and the cost-effectiveness of mitigation options is commonly assessed with abatement-cost metrics (Schäfer et al., 2016; Salgas et al., 2025). The link between the observed announcement–realisation gap and the operational and fleet levers that could compensate for it, distance band by distance band, has not been quantified.

This paper addresses that link. Its contributions are as follows.

1. A well-to-wake (WTW) emissions model disaggregated into four distance bands, in which SAF enters as a supply-constrained quantity, fleet intensity follows from a vintage model, and ATM inefficiency is derived per flight from taxi, arrival-sequencing and en-route components (Section 2).
2. A physically based base-year flight population calibrated to EUROCONTROL flight and emission shares and validated against independent seat-kilometre and per-passenger intensity data (Section 3).
3. A Shapley attribution of each pathway's reduction to levers and bands, which separates the contributions of levers that scenarios usually change together (Section 4.2).
4. A stress test of SAF-dependent pathways against supply delays and realisation shortfalls, with a closed-form critical realisation rate, and a linear programme that returns the minimum-cost corrective portfolio of ATM measures, band-targeted early retirement and alternative fuel supply (Sections 4.3 and 4.4).

We address four research questions. RQ1: How do emissions, intensity and ATM inefficiency differ across distance bands? RQ2: Which levers deliver the reductions of each pathway, in which bands? RQ3: How much SAF supply disruption can each pathway absorb before missing its targets? RQ4: What is the minimum-cost corrective action, by lever and band, and what remains uncompensable?

Section 2 presents the model, Section 3 the data, calibration and scenarios, Section 4 the results, and Section 5 the economic, management and policy implications together with the limitations. Section 6 concludes.

## 2. Model

### *2.1. Well-to-wake emissions by distance band*

Let $d \in \{1, \ldots, 4\}$ index the distance bands 0–500, 500–1500, 1500–4000 and >4000 km, and $t$ the year, with base year $t_0 = 2024$. Fuel burned by

conventional aircraft in band $d$ is

$$F_{d,t} \;=\; F^{\mathrm{eff}}_{d,t_0}\,\frac{A_{d,t}}{A_{d,t_0}}\,\Phi_{d,t}\,\big(1+a_{d,t}\big)\,\big(1-\pi_{d,t}\big), \tag{1}$$

where $F^{\mathrm{eff}}_{d,t_0}$ is base-year fuel net of ATM inefficiency, $A_{d,t}$ is activity (flights), $\Phi_{d,t}$ is the fleet fuel-intensity index ($\Phi_{d,t_0}=1$), $a_{d,t}$ is the ATM excess-fuel ratio and $\pi_{d,t}$ the share of flights operated by zero-emission propulsion (ZEP). The fuel-equivalent of ZEP flights, $R_{d,t}$, is the same expression with $\pi_{d,t}$ in place of $(1-\pi_{d,t})$. System emissions are

$$E_t \;=\; e\Big(\sum_d F_{d,t} - \theta^{\mathrm{bio}} S^{\mathrm{bio}}_t - \theta^{\mathrm{syn}} S^{\mathrm{syn}}_t\Big) \;+\; e\,\kappa_t \sum_d R_{d,t}, \tag{2}$$

where $S^{\mathrm{bio}}_t$ and $S^{\mathrm{syn}}_t$ are the quantities of bio-based and synthetic SAF used, $\theta$ their life-cycle emission reductions relative to fossil kerosene, and $\kappa_t$ the WTW intensity of the ZEP energy carrier relative to the kerosene it replaces.

*Emission boundary.* Equation (2) is a WTW inventory throughout. The fossil factor is $e = 89\,\mathrm{g\,CO_2e/MJ} \times 43.15\,\mathrm{MJ/kg} \approx 3.84\,\mathrm{t\,CO_2e}$ per tonne of fuel (ICAO, 2024), which adds upstream emissions to the tank-to-wake (TTW) combustion factor $\varepsilon = 3.16\,\mathrm{t\,CO_2/t}$. Because $\theta$ is a WTW reduction, it multiplies $e$ and never $\varepsilon$: at the tailpipe a tonne of SAF emits about as much $CO_2$ as a tonne of kerosene, and its benefit arises upstream. Applying $\theta$ to $\varepsilon$ would mix a combustion factor with a life-cycle reduction. The same boundary is applied to ZEP through $\kappa_t$; setting $\kappa_t = 0$ would credit ZEP with a full life-cycle benefit while crediting SAF only with $\theta < 1$. Two consequences follow. Indexed trajectories are unaffected by the choice between $e$ and $\varepsilon$ in the fossil term, because a constant factor cancels in a ratio; absolute tonnages are not. Inventories reported TTW, such as those of EUROCONTROL, are therefore rescaled by $e/\varepsilon$ when compared with the WTW results. Under the EU Emissions Trading System convention, eligible SAF carries a zero emission factor, which corresponds to $\theta = 1$ with $\varepsilon$ in place of $e$.

### *2.2. Intensity indicators*

Emissions per available seat-kilometre (ASK) describe the fleet; emissions per revenue passenger-kilometre (RPK) describe how efficiently passengers are carried:

$$\iota^{\mathrm{ASK}}_{d,t} = \frac{E_{d,t}}{A_{d,t}\,\bar{L}_d\,\bar{S}_d}, \qquad \iota^{\mathrm{RPK}}_{d,t} = \frac{E_{d,t}}{A_{d,t}\,\bar{L}_d\,\bar{S}_d\,\lambda_{d,t}} = \frac{\iota^{\mathrm{ASK}}_{d,t}}{\lambda_{d,t}}, \tag{3}$$

with mean stage length $\bar{L}_d$, seats $\bar{S}_d$ and load factor $\lambda_{d,t}$. Load factor enters only the RPK indicator. At flight level a higher load factor raises take-off mass and therefore fuel burn slightly, so it raises emissions per flight while lowering emissions per passenger; both effects are represented in the flight model (Section 3.1).

### *2.3. SAF mandate and supply*

ReFuelEU sets minimum shares $\sigma_t^{\min}$ and synthetic sub-shares $\sigma_t^{\text{syn}}$, defined per period and therefore stepwise in time (European Union, 2023). The obligation falls on fuel suppliers and, until 2034, may be met as a volume-weighted average across Union airports. The mandate is a fleet-wide lower bound; the upper bound is the SAF available, $\bar{S}_t$:

$$\sigma_t^{\min} \sum_d F_{d,t} \;\le\; S_t^{\text{bio}} + S_t^{\text{syn}} \;\le\; \bar{S}_t, \qquad S_t^{\text{syn}} \;\ge\; \sigma_t^{\text{syn}} \sum_d F_{d,t}. \tag{4}$$

When $\bar{S}_t$ falls below the mandated quantity, the mandate cannot be met; this is the case examined in the stress test. Equation (2) implies that for a given quantity of SAF, system emissions do not depend on how that quantity is allocated across bands:

$$\sum_d E_{d,t} \;=\; e\Big(\sum_d F_{d,t} - \theta^{\text{bio}} S_t^{\text{bio}} - \theta^{\text{syn}} S_t^{\text{syn}}\Big) + e\,\kappa_t \sum_d R_{d,t} \quad \text{for any allocation.} \tag{5}$$

In a $CO_2$ model, preferential allocation of SAF to long-haul flights is therefore abatement-neutral; arguments for it must rest on non-$CO_2$ effects, which are larger on long cruise segments (Voigt et al., 2021; Lee et al., 2021), or on cost pass-through. Band-level results below allocate SAF pro rata to fuel. A second implication matters for corrective action: when supply binds, a tonne of fuel saved by any other lever displaces fossil kerosene and abates $e$, not $e(1-\theta\sigma)$.

### *2.4. Fleet vintage and aircraft age*

Each band's fleet is represented by annual cohorts of age $0, \ldots, L-1$ with $L = 25$ years. The fuel-burn multiplier of a cohort of age $a$ relative to a current-generation aircraft is

$$m_c(a) \;=\; g_c(a)\,(1+\rho\,a), \qquad g_c(a) = \begin{cases} 1 & a \le \bar{a}_c \\ (1-\Delta_c)^{-1} & a > \bar{a}_c \end{cases}, \tag{6}$$

for aircraft class $c$ (regional, narrow-body, wide-body), with generation step $\Delta_c$, current-generation age limit $\bar{a}_c$ and deterioration $\rho$. Cohorts retire at age $L$ and are replaced by deliveries whose multiplier falls by a factor $(1-\tau)$ per year. Accelerated renewal retires the oldest cohorts early, up to a delivery-slot limit expressed as a share of the band's fleet. $\Phi_{d,t}$ is the kilometre-weighted mean multiplier divided by its base-year value.

This structure answers how aircraft age enters the analysis. Because age acts multiplicatively on fuel burn, the natural regression on flight-level data is

$$\ln y_i \;=\; \beta_0 + \sum_{d=2}^{4} \beta_d B_{id} + \sum_{d=1}^{4} \gamma_d \, B_{id} A_i + \eta \, \Lambda_i + \mathbf{x}_i' \boldsymbol{\zeta} + \epsilon_i, \tag{7}$$

with band indicators $B_{id}$, age $A_i$, load factor $\Lambda_i$ and controls $\mathbf{x}_i$ (log distance, aircraft class). Each $\gamma_d$ is a semi-elasticity: the *proportional* change in emissions per year of age in band $d$. The *absolute* change is $\partial E_i / \partial A_i = \gamma_d E_i$, which grows with the band's emission level even when the $\gamma_d$ are equal. A common slope $\gamma$ therefore already implies a much larger absolute age effect at long range; whether the relative effect also differs is tested by $H_0 \colon \gamma_4 = \gamma_1$. The load-factor effect is estimated twice, with $y_i$ equal to emissions per flight and to emissions per RPK.

### *2.5. Air traffic management inefficiency*

ATM-related excess fuel of flight $i$ is the sum of additional taxi-out time, additional arrival-sequencing (ASMA) time and en-route inefficiency within European airspace:

$$x_i \;=\; \tau^{\text{taxi}} q^{\text{taxi}}_{c(i)} \;+\; \tau^{\text{asma}} q^{\text{hold}}_{c(i)} \;+\; \left(k^{\text{KEA}} + k^{\text{oth}}\right) C_i \, \frac{\min(D_i, D^{\text{EU}})}{D_i}, \tag{8}$$

where $\tau$ are additional times, $q$ class-specific fuel flows, $k^{\text{KEA}}$ the horizontal en-route flight-efficiency excess, $k^{\text{oth}}$ the remaining (vertical and other) inefficiency, $C_i$ cruise fuel, $D_i$ stage length and $D^{\text{EU}}$ the distance flown in European airspace. The two fixed terminal terms make the relative inefficiency largest on short flights, and the airspace limit makes it smallest on long-haul flights. The band ratio $a_{d,t_0}$ in Eq. (1) is the band's excess divided by its efficient fuel. ATM measures recover a fraction of this pool that rises linearly to a ceiling by a target year; the pool cannot be fully recovered, because part of the inefficiency is needed for separation and weather (Performance Review Commission, 2021).

### 2.6. Zero-emission propulsion

Battery-electric and hydrogen aircraft are limited by range (Akinola et al., 2025; Uddin and Wang, 2025). Penetration is bounded by band-specific ceilings, $0 \le \pi_{d,t} \le \bar{\pi}_d$ with $\bar{\pi}_1 > \bar{\pi}_2 > \bar{\pi}_3 = \bar{\pi}_4 = 0$, and ramps up from an entry-into-service year.

### 2.7. SAF supply disruption and tolerance

Supply disruption is parameterised by a ramp-up delay $\delta$ (years) and a persistent realisation rate $r \in (0, 1]$:

$$S_t(\delta, r) \;=\; r\, \sigma^{\min}_{t-\delta}\, F^{\mathrm{nom}}_t, \tag{9}$$

applied separately to the bio and synthetic components, where $F^{\mathrm{nom}}_t$ is the pathway's nominal fuel demand. A delay shifts the capacity ramp in time; $r < 1$ represents capacity that never materialises. The evidence in Martulli et al. (2025) bounds $r$ to roughly $[0.57, 0.95]$.

For each pathway $p$ we consider two targets: the cumulative 2025–2050 budget $B_p = \sum_t E^{p,\mathrm{nom}}_t$ and the 2050 level $E^{p,\mathrm{nom}}_{2050}$, each with an optional slack $s \ge 0$. For a shortfall without delay, Eq. (2) gives the critical realisation rate for the 2050 target in closed form:

$$r^*_p \;=\; \frac{F_{2050} + \kappa_{2050} R_{2050} - \bar{E}_p/e}{\theta^{\mathrm{bio}} S^{\mathrm{bio}}_{2050} + \theta^{\mathrm{syn}} S^{\mathrm{syn}}_{2050}}, \tag{10}$$

with $\bar{E}_p = (1+s) E^{p,\mathrm{nom}}_{2050}$ for an internal target or a level common to all pathways for an external one. For internal targets and $\kappa_{2050} = 0$, $r^*_p = 1 - s\,(1/(\bar{\theta}\sigma^{\min}_{2050}) - 1)$, so tolerance depends only on the terminal blend and its life-cycle reduction; efficiency levers matter for robustness only against external targets.

### 2.8. Minimum corrective action

When a disruption breaks a target, three levers can compensate. Alternative SAF $x^A_t$ (imports or book-and-claim supply) abates $e\,\theta^{\mathrm{bio}}$ per tonne and is limited to a share of annual fuel demand. Early retirement of $x^R_{d,t}$ aircraft in band $d$ saves $f_{d,t}$ tonnes of fuel per year for the $R$ years by which retirement is advanced, limited by the delivery slots the pathway does not already use. Additional ATM measures save $x^O_{d,t}$ tonnes of fuel, limited by the part of the benefit pool between the nominal recovery and an upper ceiling. With

$G_t = E_t(\delta, r) - E_t^{\mathrm{nom}}$ the gap created by the disruption, the minimum-cost corrective portfolio solves

$$\begin{aligned}
\min\ & \sum_t c^A x_t^A + \sum_{d,t} C_{d,t}^R x_{d,t}^R + \sum_{d,t} c^O x_{d,t}^O + M\big(u^{\mathrm{cum}} + u^{2050}\big) \qquad (11)\\
\text{s.t.}\ & \sum_t \Big[ G_t - e\theta^{\mathrm{bio}} x_t^A - e \sum_d \Big( \sum_{\tau=t-R+1}^{t} f_{d,\tau} x_{d,\tau}^R + x_{d,t}^O \Big) \Big] \le u^{\mathrm{cum}},\\
& G_{2050} - e\theta^{\mathrm{bio}} x_{2050}^A - e \sum_d \Big( \sum_{\tau=2051-R}^{2050} f_{d,\tau} x_{d,\tau}^R + x_{d,2050}^O \Big) \le u^{2050},\\
& 0 \le x_t^A \le \bar{x}_t^A, \quad 0 \le x_{d,t}^R \le \bar{x}_{d,t}^R, \quad 0 \le x_{d,t}^O \le \bar{x}_{d,t}^O, \quad u \ge 0.
\end{aligned}$$

Here $c^A$ is the SAF price premium over fossil fuel, $C_{d,t}^R$ the net cost of advancing one retirement and $c^O$ the net cost per tonne of fuel saved by ATM measures. The penalty $M$ makes the residuals $u$ positive only when the levers cannot close the gap within their limits; a positive residual is the part of the disruption that requires measures outside the model, such as demand management or carbon removal. Measures with negative net cost would already be adopted in the nominal pathway unless barriers prevented it, so in the objective their cost is set to zero plus a small charge per tonne abated (the same for every lever), ordered by their net cost; they are therefore used only where needed, and their true net cost is reported. The static abatement cost of advancing one retirement in band $d$ is

$$\mu_d \;=\; \frac{\mathrm{CRF}\,\big(K_d - V_d\big) - P^{\mathrm{fuel}} f_d}{e\, f_d}, \qquad f_d = U_d\, \phi_d\, \big(m_d^{\mathrm{old}} - m_d^{\mathrm{new}}\big)\big(1 - \pi_d\big), \qquad (12)$$

with new-aircraft value $K_d$, residual value $V_d$, capital recovery factor CRF, fuel price $P^{\mathrm{fuel}}$, utilisation $U_d$ (km per year) and fuel burn per kilometre $\phi_d$ (scaled by the share $1 - \pi_d$ of the band still flown on kerosene), following the abatement-cost logic of Schäfer et al. (2016) and Salgas et al. (2025). Abatement per aircraft is proportional to $f_d$ and favours long-haul aircraft; abatement per euro also depends on the capital outlay $K_d - V_d$, and the ranking is an empirical result, not an assumption.

## 3. Data, calibration and scenarios

### *3.1. Flight-level fuel model*

No open flight-level emission inventory with aircraft age and load factor exists for Europe. The base-year population is therefore generated from a physically based fuel model, and every downstream result can be recomputed on observed flights with the same schema. We sample 60,000 flights with the 2019 EUROCONTROL band shares of departures (EUROCONTROL, 2021; Performance Review Commission, 2021), draw stage lengths within each band, assign an aircraft class from band-specific class mixes, and draw age and load factor. Fuel per flight is

$$F_i \;=\; \left[\ell_c + \kappa_c\,\phi_c^0\,D_i\,(1+\omega_c D_i)\right] m_c(A_i)\left[1+\psi(\Lambda_i-\bar{\Lambda})\right] \;+\; x_i, \qquad (13)$$

with landing-and-take-off fuel $\ell_c$, current-generation cruise fuel per kilometre $\phi_c^0$, a fuel-weight curvature $\omega_c$, the vintage multiplier of Eq. (6), payload sensitivity $\psi$ and the ATM excess of Eq. (8). Band differences therefore arise from distance, aircraft size and fleet composition rather than from assumed band means.

### *3.2. Calibration and validation*

Two class scalings, $\kappa_{\mathrm{RG}}$ and $\kappa_{\mathrm{WB}}$, are fitted so that band $CO_2$ shares reproduce the EUROCONTROL 2019 shares; the narrow-body scaling is fixed at one and anchors the level. The remaining en-route inefficiency $k^{\mathrm{oth}}$ is fitted so that the ATM benefit pool of flights below 4000 km equals the 9% reported by the Performance Review Commission (Performance Review Commission, 2025). The fitted scalings are $\kappa_{\mathrm{RG}} = 1.01$ and $\kappa_{\mathrm{WB}} = 1.14$, close to the engineering start values, and $k^{\mathrm{oth}} = 6.7\%$. The largest relative deviation from the calibration targets is 4.7%. The band totals are then scaled to the 202.7 Mt $CO_2$ (TTW) emitted in the EUROCONTROL area in 2024 (Performance Review Commission, 2025).

Three checks use data not used in the fit (Table 1). The long-haul band carries 10.4% of seats and 41.2% of seat-kilometres, against 10% and "more than 40%" reported by EUROCONTROL (2021), and emissions per RPK follow the U-shaped pattern across bands found in independent German emission factors (ifeu, 2026). As a comparison with an external projection, the reference pathway without any improvement of new deliveries reaches 267 Mt $CO_2$ (TTW) in 2050, against 271 Mt in EUROCONTROL's case without significant measures (EUROCONTROL, 2024); with the assumed 1% annual

improvement of new deliveries it reaches 232 Mt, an improvement that the outlook counts among its fleet and technology measures.

Table 1: Calibration and validation of the base-year flight population. Flight and $CO_2$ shares in %; intensities in g $CO_2$ (TTW). Targets for flight and $CO_2$ shares from EUROCONTROL (2021) and Performance Review Commission (2021); the per-RPK ratio is compared with ifeu (2026), which was not used in the fit.

| | Flights | | $CO_2$ | | Intensity | | RPK ratio | | ATM |
|---|---|---|---|---|---|---|---|---|---|
| Band | target | model | target | model | /ASK | /RPK | model | ifeu | excess % |
| 0-500 km | 24.0 | 23.9 | 3.8 | 3.8 | 107 | 131 | 1.00 | 1.00 | 12.0 |
| 500-1500 km | 46.0 | 46.2 | 21.2 | 20.2 | 72 | 88 | 0.68 | 0.69 | 10.3 |
| 1500-4000 km | 24.0 | 24.1 | 27.0 | 28.3 | 66 | 80 | 0.62 | 0.55 | 7.7 |
| >4000 km | 6.0 | 5.7 | 48.0 | 47.7 | 92 | 112 | 0.86 | 0.73 | 2.7 |

### *3.3. Scenarios*

Four nested pathways are defined by the levers they activate (Table 2). All share traffic growth of 1.6% per year (EUROCONTROL, 2024) and natural fleet renewal with a 1% annual improvement of new deliveries. S0 is a counterfactual without new measures; it holds the SAF share at its 2024 level of 0.6% (EASA, 2025b) and is therefore below the legal mandate. S1 adds ATM measures that recover half of the benefit pool by 2035. S2 adds the ReFuelEU mandate, fully supplied. S3 adds zero-emission propulsion from 2035 and accelerated renewal that lowers the retirement age to 20 years within the delivery-slot limit. All parameters, their sources and their status (sourced, derived, calibrated, assumed) are listed in Appendix A.

Table 2: Scenario definitions. All scenarios include traffic growth and natural fleet renewal.

| | Pathway | ATM measures | SAF mandate | ZEP propulsion | Accelerated renewal |
|---|---|---|---|---|---|
| S0 | Reference | – | – | – | – |
| S1 | Operational efficiency | ✓ | – | – | – |
| S2 | SAF mandate | ✓ | ✓ | – | – |
| S3 | Low-emission propulsion | ✓ | ✓ | ✓ | ✓ |

## 4. Results

### *4.1. Band structure and ATM inefficiency*

Figure 1 shows where flights, seat-kilometres and emissions sit. On average a long-haul flight emits about 52 times as much as a flight below 500 km. Most of this difference reflects distance and aircraft size, not inefficiency. Per seat-kilometre the long-haul band emits 92 g, against 72 g and 66 g in the two medium bands and 107 g below 500 km, where take-off and landing dominate. The concentration of emissions in long-haul flying therefore describes where abatement must reach, not that long-haul flying is intrinsically wasteful.

ATM inefficiency has the opposite profile. Excess fuel is 12.0% of fuel below 500 km, 10.3% and 7.7% in the medium bands and 2.7% above 4000 km, because terminal inefficiency is a fixed cost per flight and only the European part of a long-haul trajectory is affected. ATM measures are thus a short-haul and medium-haul lever, complementary to fleet renewal, whose leverage grows with distance.

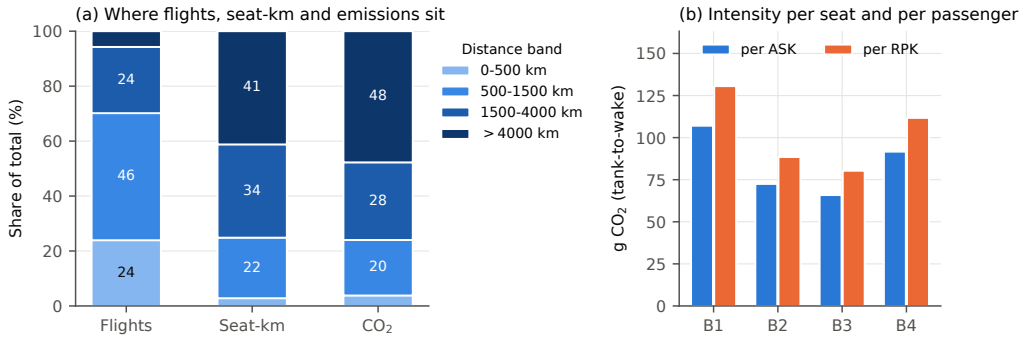



Figure 1: Base-year band structure. (a) Shares of flights, seat-kilometres and $CO_2$ by band. (b) Emissions per available seat-kilometre and per revenue passenger-kilometre (TTW).

The regression of Eq. (7) on the generated flights returns age semi-elasticities of 0.95, 1.12, 1.18 and 1.66% per year for bands 1 to 4 (standard errors of at most 0.02 percentage points; $H_0 : \gamma_4 = \gamma_1$ rejected, $F = 1477$). The relative gradient is steeper at long range because the assumed generation step is larger for wide-bodies than for regional aircraft ($\Delta_{\mathrm{WB}} = 22\%$ against 15%); on generated data this coefficient reproduces an input and is reported only to show what Eq. (7) identifies. The robust result is the absolute one: one year of age adds about 94 times more $CO_2$ to a long-haul flight than to a

flight below 500 km. The load-factor coefficient is positive for emissions per flight ($\eta = 0.15$) and negative for emissions per RPK ($\eta = -1.10$), as Eq. (3) requires.

### *4.2. Pathways and attribution*

Figure 2(a) shows the WTW emission index. Without new measures (S0), emissions rise to an index of 115 in 2050, as traffic growth by a factor of 1.51 outweighs the fleet-intensity improvement to 0.76. ATM measures (S1) lower the 2050 index to 111. The SAF mandate (S2) reduces it to 49, and the full portfolio (S3) to 48. The S2 and S3 trajectories fall in steps that mirror the periods of the mandate, most sharply in 2050.

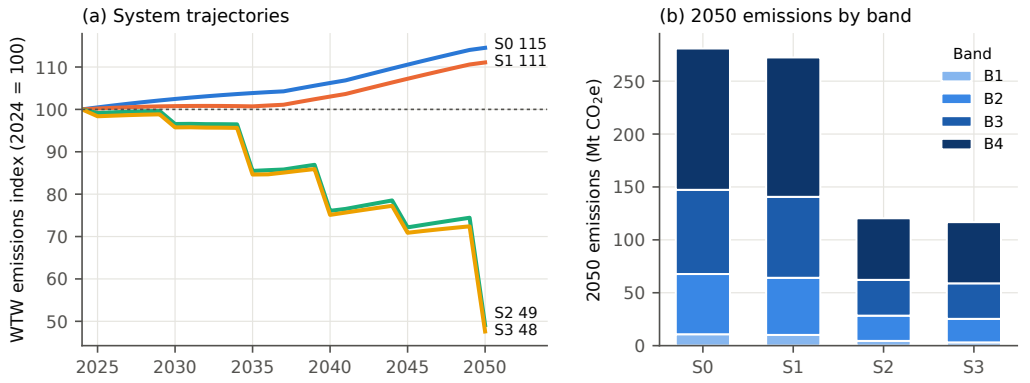



Figure 2: (a) Well-to-wake emission index by pathway (2024 = 100); the steps in S2 and S3 follow the ReFuelEU periods. (b) 2050 emissions by distance band.

Because scenarios activate several levers at once, differences between them do not identify the contribution of any one lever. Figure 3 reports the Shapley attribution of the reduction from S0 to the full lever set, by lever and band. Of the 164 Mt reduction in 2050, the SAF mandate contributes 92%, ATM measures 4%, zero-emission propulsion 3% and accelerated renewal 1%. Accelerated renewal contributes little in 2050 because natural turnover has replaced the 2024 fleet by then; its effect is cumulative and appears in panel (b). Over 2025–2050 the shares are 84% for SAF, 10% for ATM, 3% for ZEP and 2% for renewal. Long-haul flights receive 49% of the SAF contribution, while the whole ZEP contribution falls in the two shortest bands. The difference between S3 and S2 is therefore small, and attributing it to propulsion technology without a decomposition would overstate the role of ZEP.

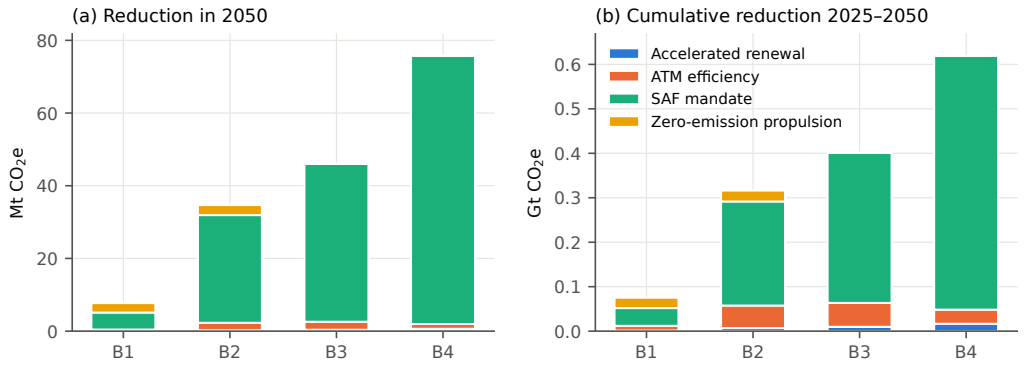



Figure 3: Shapley attribution of the reduction from the reference (S0) to the full lever set, by lever and distance band. (a) 2050 emissions. (b) Cumulative 2025–2050 emissions.

### *4.3. Tolerance to SAF supply disruption*

Figure 4 maps the cumulative excess emissions of S2 and S3 over the $(\delta, r)$ plane. Within the range reported by Martulli et al. (2025), a persistent 18% shortfall ($r = 0.82$) raises cumulative 2025–2050 emissions of S2 by 4.5%; combined with a three-year delay the excess is 9.0%, and the 2050 level rises by 67%. The two targets react very differently to delay. Because the mandate steps from 42% to 70% on 1 January 2050, a one-year delay raises S2's 2050 emissions by 52% but its cumulative emissions by only 2.7%. A target set on the terminal year therefore measures the timing of the last mandate step more than the climate effect of the pathway.

For persistent shortfalls the critical realisation rate of Eq. (10) is 0.96, 0.92 and 0.84 for slacks of 5, 10 and 20% on S2's own 2050 level, and identical to two decimals for S3. With $\kappa_{2050} = 0$ the closed form makes them exactly equal; the upstream emissions of S3's zero-emission flights ($\kappa_{2050} = 0.10$) change $r^*$ by less than 0.01. Pathways that differ in efficiency but share the terminal blend are therefore equally fragile against their own targets. Against an external target common to both, set 10% above S2's nominal 2050 level, the critical rate is 0.92 for S2 and 0.89 for S3: the additional fuel savings of S3 buy robustness only when the target is fixed in absolute terms.

### *4.4. Minimum corrective action*

Table 3 reports the static abatement cost of each corrective lever. ATM measures have a negative net cost (EUR -68 per tonne $CO_2$e), because the fuel they save is worth more than their assumed implementation cost. Advancing

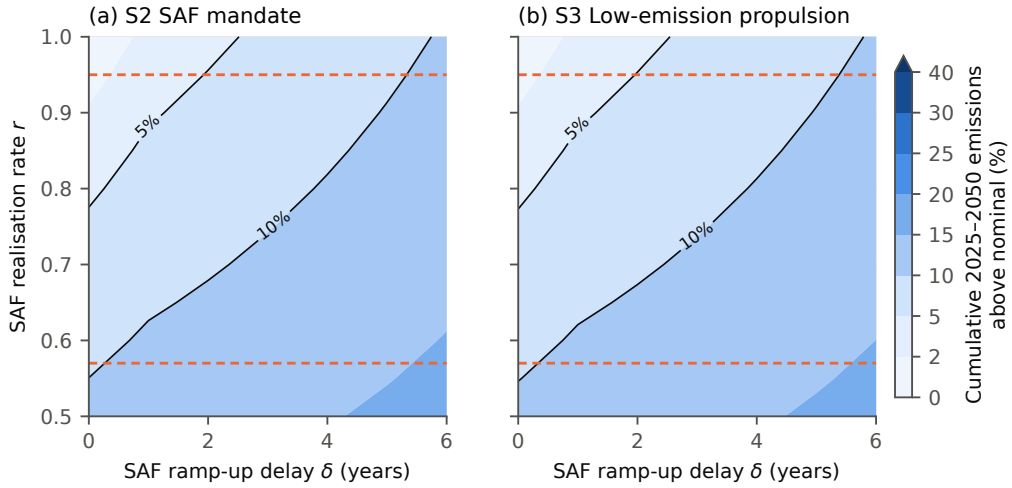



Figure 4: Cumulative 2025–2050 emissions above the nominal pathway under SAF supply disruption, by ramp-up delay $\delta$ and realisation rate $r$. Dashed lines bound the range of realisation rates implied by Martulli et al. (2025).

the retirement of a long-haul aircraft costs EUR 10 per tonne and abates 55 kt $CO_2$e per year, against EUR 261–272 per tonne and 5.0–9.5 kt per year in the other bands. Fuel savings grow faster with distance than the capital outlay, so long-haul renewal is cheapest per tonne as well as largest per aircraft under the stated price assumptions. Alternative bio-SAF costs EUR 469 and synthetic SAF EUR 2132 per tonne at the 2024 ReFuelEU reference prices (EASA, 2025a).

Table 3: Static abatement cost of corrective levers (EUR per t $CO_2$e, WTW, 2030) and annual abatement per early-retired aircraft.

| Lever | EUR/t $CO_2$e | kt $CO_2$e per aircraft-year |
|---|---|---|
| Accelerated renewal, 0-500 km | 261 | 5.0 |
| Accelerated renewal, 500-1500 km | 262 | 8.5 |
| Accelerated renewal, 1500-4000 km | 272 | 9.5 |
| Accelerated renewal, >4000 km | 10 | 54.7 |
| ATM efficiency (all bands) | -68 | – |
| Alternative bio-SAF | 469 | – |
| Alternative synthetic SAF | 2132 | – |

Table 4 and Figure 5 report the solutions of programme (11) for four reference disruptions, first with only the cumulative budget as target and then with both targets. For S2 under an 18% shortfall without delay, the cumulative gap is 217 Mt $CO_2$e. Restoring the budget alone requires ATM measures and the early retirement of 297 long-haul aircraft and yields a net saving of EUR 7.1 bn, because under the assumed fuel and aircraft prices both levers save more fuel than they cost. Restoring the 2050 level as well raises the cost to EUR 17.2 bn: the 2050 constraint can only be met by actions that save fuel in 2050 itself, which requires renewal in the other bands and alternative SAF, and 3% of the 2050 gap still remains. Restricted to a single lever, ATM measures close only 55% of the cumulative gap, whereas renewal alone or alternative SAF alone can close it. Under a three-year delay with the same shortfall, the cumulative gap grows to 473 Mt and costs EUR 71 bn to close (EUR 77 bn with both targets), and 66% of the 2050 gap cannot be closed within the delivery-slot and supply limits.

Table 4: Minimum-cost corrective portfolios. Gaps in Mt $CO_2$e (cumulative 2025–2050 and in 2050). Lever quantities refer to the programme with both targets: ATM in Mt fuel saved; aircraft retired early (cumulative); alternative SAF in Mt. Net cost in bn EUR when only the cumulative budget is restored and when both targets are pursued; negative values are net savings. Residuals are the uncompensated shares of the cumulative and 2050 gaps (%). Disruption given as $(\delta, r)$.

| | | Gap | | ATM | Early retirement | | Alt. | Cost | | Residual | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Path | $(\delta, r)$ | cum. | 2050 | fuel | >4000 km | other | SAF | cum. | both | cum. | 2050 |
| S2 | (3, 1.00) | 311 | 62.9 | 31.2 | 562 | 1,422 | 3.6 | 10.8 | 25.5 | 0 | 57 |
| S2 | (0, 0.82) | 217 | 27.6 | 31.2 | 305 | 993 | 3.6 | -7.1 | 17.2 | 0 | 3 |
| S2 | (3, 0.82) | 473 | 79.2 | 31.2 | 698 | 3,410 | 17.1 | 71.0 | 76.6 | 0 | 66 |
| S2 | (3, 0.65) | 625 | 94.5 | 31.2 | 698 | 3,410 | 70.0 | 142.4 | 148.1 | 0 | 72 |
| S3 | (3, 1.00) | 305 | 60.6 | 30.6 | 15 | 69 | 63.0 | 78.2 | 78.4 | 0 | 78 |
| S3 | (0, 0.82) | 212 | 26.6 | 30.6 | 15 | 69 | 30.8 | 34.8 | 34.9 | 0 | 50 |
| S3 | (3, 0.82) | 463 | 76.2 | 30.6 | 15 | 78 | 85.0 | 108.3 | 108.3 | 20 | 83 |
| S3 | (3, 0.65) | 612 | 91.0 | 30.6 | 15 | 78 | 85.0 | 108.3 | 108.3 | 40 | 85 |

S3 is less correctable than S2. Its own accelerated renewal already uses almost all delivery slots, so the renewal lever is nearly exhausted when a disruption occurs. The corrective programme must then rely on alternative SAF, which is more expensive: restoring the cumulative budget after an 18% shortfall costs EUR 34.8 bn, and 50% of the 2050 gap remains open even without delay. When a three-year delay is combined with a shortfall of 18%

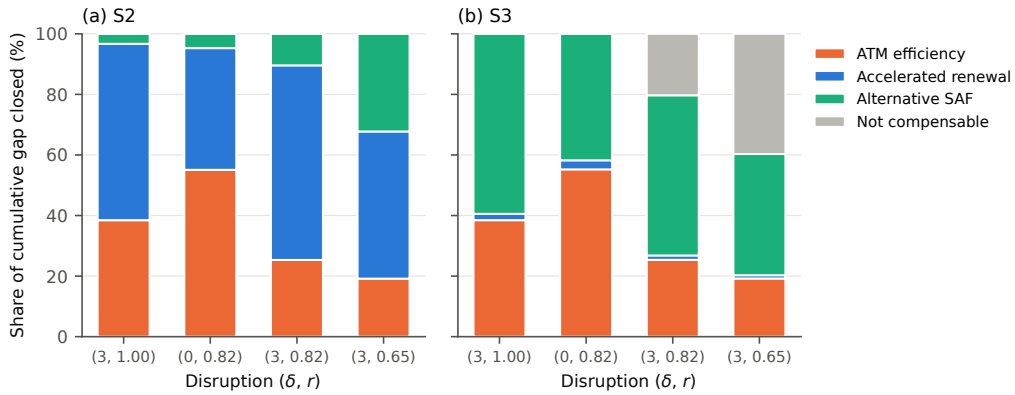



Figure 5: Composition of the minimum-cost corrective portfolio (both targets) as a share of the cumulative 2025–2050 gap. Grey: part of the cumulative gap that cannot be closed within the lever limits.

or more, up to 40% of the cumulative gap also remains uncompensated. A pathway that pre-commits a scarce resource to its nominal plan has less of it left as a reserve.

### *4.5. Sensitivity*

Figure 6 varies eight parameters one at a time for S2 under an 18% shortfall with both targets. To keep outcomes comparable, any residual the levers cannot close, cumulative or in 2050, is priced at the synthetic-SAF abatement cost. The resulting corrective cost of EUR 18.8 bn is most sensitive to the implementation cost of ATM measures and to the fossil fuel price, both of which determine the net cost of fuel savings; the widest range is EUR 32 bn. The life-cycle reduction of bio-SAF barely matters, because it scales the gap and the abatement of alternative SAF alike. Long-haul renewal remains the cheapest renewal option in every variation. ATM measures remain cheaper than long-haul renewal as long as their gross implementation cost stays below about EUR 245 per tonne of $CO_2$; at the high value of the range the two swap places.

## 5. Economic, management and policy implications

### *5.1. ATM as the first buffer against SAF shortfalls*

ATM measures have a negative net cost under the stated assumptions, and their benefit is concentrated in the short- and medium-haul bands where

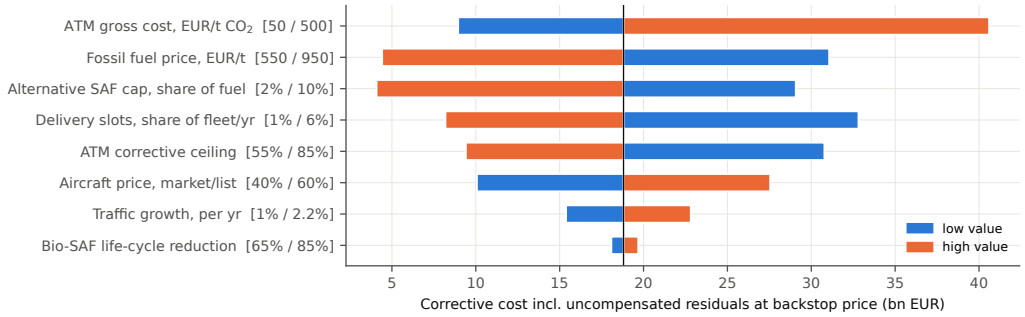



Figure 6: One-at-a-time sensitivity of the corrective cost (both targets) for S2 under an 18% shortfall without delay, with uncompensated residuals priced at the synthetic-SAF abatement cost. Low and high values in brackets.

fleet renewal has the least leverage. They are, however, small: even at a recovery ceiling of 70% of the benefit pool, they close only about half of the cumulative gap from an 18% shortfall. The model lets every lever act from 2025; in practice ATM measures can also be deployed faster than fleet renewal, which is constrained by manufacturing lead times, so the model understates their value as a near-term buffer. Two implications follow for performance regulation under the Single European Sky. First, environmental targets for air navigation service providers can be framed as a contingency instrument: the benefit pool not yet recovered is a reserve that can be mobilised when fuel supply falls short. Second, because the pool is largest in relative terms on short flights, terminal-area measures (taxi and arrival management) deserve weight comparable to en-route flight efficiency, which dominates the current indicators.

### *5.2. SAF target design*

The stepwise mandate makes the 2050 level hostage to the timing of the last step. A one-year slip changes 2050 emissions by about half but cumulative emissions by a few percent. Monitoring progress against a cumulative budget, or against interim milestones between the five-year steps, would align the assessment with the physical climate effect and reduce the incentive to judge pathways by a single year. For fuel suppliers and airlines, the results quantify the value of contingency arrangements, such as import options and book-and-claim supply, that can be activated when domestic capacity is late.

### 5.3. Fleet planning and corrective headroom

Per aircraft and per euro, advancing the retirement of long-haul aircraft is the most effective renewal measure, so fleet-renewal support aimed at emissions should not be distributed evenly across the network. Delivery slots, however, are scarce. A pathway that commits them in advance to accelerated renewal reduces its emissions but loses the ability to respond when SAF falls short. For airlines and policymakers this is a portfolio question: part of the renewal capacity has option value as a reserve, and committing it fully trades robustness for expected reductions.

### 5.4. Who bears the corrective cost

The levers belong to different actors. ATM measures are delivered by air navigation service providers and the Network Manager under performance regulation; early retirement is an airline decision constrained by manufacturers' backlogs; alternative SAF is procured by fuel suppliers, whose shortfalls are penalised with reference to the published prices. The corrective portfolios in Table 4 therefore require coordination. In S2 the least-cost response to a supply shortfall lies mostly with air navigation service providers and airlines rather than with the fuel suppliers who bear the mandate. In S3, where renewal capacity is already committed, alternative SAF procured by fuel suppliers closes 40–59% of the cumulative gap.

### 5.5. Limitations

Four limitations qualify the results. First, the base-year flight population is generated rather than observed. It reproduces the published band shares and passes independent checks, but the within-band regressions on generated data reproduce their inputs and are not evidence; the pipeline accepts observed flights with the same schema, and replication on such data is the most important extension. Second, the model is restricted to $CO_2$ on a WTW basis. Non-$CO_2$ effects, contrails in particular, are of similar magnitude (Lee et al., 2021), are distance- and altitude-dependent, and would change both the allocation question of Eq. (5) and the value of ATM measures, which can be used to avoid contrail formation. Third, costs are static and undiscounted over the corrective window, demand does not respond to cost pass-through, and the corrective programme abstracts from aircraft reassignment across markets. Fourth, the stress test is a deterministic grid; attaching probabilities to $(\delta, r)$, for example from the diffusion scenarios of Martulli et al. (2025), would convert the tolerance maps into probabilities of

meeting each target. The geographic scopes also differ: the inventory covers the EUROCONTROL area, whereas ReFuelEU applies at Union airports.

## 6. Conclusions

This paper developed a well-to-wake, distance-band model of European aviation emissions and used it to ask how much SAF supply shortfall decarbonisation pathways can absorb and how it can be compensated at least cost. Four results stand out. First, emissions concentrate in long-haul flying while ATM inefficiency concentrates in short-haul flying, so the levers that can act quickly and those that act where emissions are largest belong to different bands. Second, the SAF mandate delivers most of the reduction of the full portfolio, and the step structure of the mandate makes terminal-year targets far more sensitive to supply timing than cumulative budgets. Third, after a supply shortfall the cumulative budget can be restored mainly through ATM measures and early retirement of long-haul aircraft, at low or negative net cost under the assumed prices; restoring the 2050 level as well is considerably more expensive, and under multi-year delays most of the 2050 gap cannot be closed within manufacturing and supply limits. Fourth, pathways that commit scarce delivery slots in advance retain the least corrective headroom.

For policy, the results support assessing ReFuelEU progress against cumulative budgets or interim milestones, treating the unrecovered ATM benefit pool as a contingency reserve, and directing fleet-renewal support to long-haul aircraft while preserving part of the renewal capacity as an option. Future work should replicate the analysis on observed flight-level data, add non-$CO_2$ effects, and attach probabilities to the disruption space.

## Data and code availability

The model, parameter file with sources, and the script that reproduces every number, table and figure in this paper are available in a public repository [repository URL].

## Declaration of generative AI use

## CRediT authorship contribution statement

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Appendix A. Parameters and sources

| Parameter | Value | Status | Source or rationale |
|---|---|---|---|
| `emission_factors.ttw_co2_per_t_fuel` | 3.16 | sourced | ICAO CO2 factor for Jet A-1 (used by EUROCONTROL and EASA) |
| `emission_factors.wtw_fossil_g_per_mj` | 89.0 | sourced | CORSIA default life-cycle value for fossil jet fuel (icao_corsia) |
| `emission_factors.lhv_mj_per_kg` | 43.15 | sourced | ICAO / CORSIA lower heating value of jet fuel |
| `bands.stage_triangular_km` | see repository config | assumption | shape chosen to reflect typical European route structure; band edges from eurocontrol2021 |
| `bands.flight_share_2019` | 0.24, 0.46, 0.24, 0.06 | sourced | eurocontrol2021 (24 % below 500 km, 6 % above 4000 km) and prc2021 (about 70 % below 1500 km) |
| `bands.co2_share_2019` | 0.038, 0.212, 0.27, 0.48 | sourced | eurocontrol2021 (3.8 % and 48 %) and prc2021 (about 25 % below 1500 km) |
| `bands.seatkm_share_longhaul_min` | 0.4 | validation | eurocontrol2021 ("more than 40 % of seat-kilometres") |
| `bands.seat_share_longhaul` | 0.1 | validation | eurocontrol2021 ("10 % of total seats") |
| `bands.gco2e_per_pkm_germany_2024` | 284, 195, 156, 207 | validation | ifeu2026 (TREMOD-AV, German departures, average cabin, incl. WTT); compared as ratios only |
| `aircraft_classes.seats` | 76, 180, 320 | assumption | typical European configurations (ATR72/E175, A320/737-800, A330/A350/787) |
| `aircraft_classes.lto_fuel_t` | 0.25, 0.6, 1.8 | assumption | order of magnitude of ICAO LTO-cycle fuel for representative types (ICAO Engine Emissions Databank) |
| `aircraft_classes.cruise_fuel_kg_per_km` | 1.5, 2.4, 5.6 | calibrated | current-generation type values (ATR72-600/E175, A320neo, A350-900) of order 1.5/2.4/5.6 kg per km; RG and WB scaled by the code to match co2_share_2019, NB fixed as level anchor |
| `aircraft_classes.weight_curvature_per_km` | 0.0, 2e-05, 2e-05 | assumption | carriage of fuel weight raises fuel per km with range (Breguet); calibrated scaling absorbs level |

| Parameter | Value | Status | Source or rationale |
|---|---|---|---|
| aircraft_classes.taxi_fuel_kg_per_min | 6.0, 11.0, 22.0 | assumption | ICAO idle (7 % thrust) fuel flow, order of magnitude for representative engines |
| aircraft_classes.hold_fuel_kg_per_min | 22.0, 40.0, 95.0 | assumption | low-altitude holding fuel flow, order of magnitude (BADA-type values) |
| aircraft_classes.utilisation_km_per_year | 1.0e6, 2.3e6, 4.6e6 | assumption | about 2 000 / 3 200 / 4 800 block hours per year at typical block speeds |
| aircraft_classes.newgen_fuel_reduction | 0.15, 0.17, 0.22 | assumption | ATR-600 / A320neo-737MAX / A350-787 vs predecessor generation, manufacturer claims of 15-25 % |
| aircraft_classes.newgen_max_age_years | 6, 8, 12 | assumption | entry into service of current generation (2016-2018 NB, 2011-2015 WB) relative to base year |
| aircraft_classes.list_price_musd | 27.0, 110.6, 317.4 | sourced | Airbus 2018 list prices (A320neo, A350-900); ATR72-600 widely reported list price |
| aircraft_classes.market_to_list_ratio | 0.5 | assumption | typical discount of transaction to list price; varied 0.4-0.6 |
| aircraft_classes.residual_value_share_old | 0.2 | assumption | residual value of an aircraft retired near end of economic life, as share of new market value |
| aircraft_classes.usd_to_eur | 0.92 | assumption | average 2024 exchange rate |
| band_class_mix | see repository config | assumption | share of flights by aircraft class in each band; varied in sensitivity |
| fleet.age_max_years | 25 | assumption | typical retirement age of commercial aircraft in Europe |
| fleet.deterioration_per_year | 0.002 | assumption | engine/airframe performance deterioration of about 0.1-0.3 % per year |
| fleet.new_delivery_improvement_per_year | 0.01 | assumption | fuel-efficiency improvement of successive deliveries, below the ICAO 2 %/yr aspirational goal |
| fleet.load_factor_mean | 0.82 | assumption | European passenger load factor around 80-85 % (2019 and 2024) |
| fleet.payload_fuel_sensitivity | 0.15 | assumption | relative fuel increase per unit increase in load factor (payload weight about 10-15 % of take-off mass) |
| demand.flights_growth_per_year | 0.016 | sourced | eurocontrol2024outlook base scenario, average annual growth 2023-2050 |
| demand.growth_low_high | 0.01, 0.022 | assumption | range for sensitivity, bracketing the outlook's low and high scenarios |
| demand.co2_2024_mt | 202.7 | sourced | prr2024 (CO2 of flights in the EUROCONTROL area, 2024, tank-to-wake) |
| atm.additional_taxi_out_min | 4.0 | assumption | network-average additional taxi-out time; airport values in prr2024 range 2-8 min |
| atm.additional_asma_min | 2.0 | assumption | network-average additional ASMA time; T30 airport values in prr2024 up to 7.5 min |
| atm.kea_excess_distance | 0.0316 | sourced | prr2024 horizontal en-route flight efficiency of the actual trajectory, 2024 |
| atm.european_airspace_km | 2000 | assumption | share of a long-haul trajectory flown in EUROCONTROL airspace; en-route inefficiency applied to min(d, this) |

| Parameter | Value | Status | Source or rationale |
|---|---|---|---|
| atm.benefit_pool_intra_europe | 0.09 | sourced | prr2024 ("ATM-related benefit pool ... around 9 %"); calibration target for flights below 4000 km |
| atm.recoverable_fraction | 0.5 | assumption | prc2021 notes the pool cannot be fully recovered; varied 0.3-0.7 |
| atm.recovery_full_year | 2035 | assumption | horizon for SES/SESAR deployment in the operational-efficiency scenario |
| atm.corrective_max_fraction | 0.7 | assumption | upper bound on the share of the benefit pool recoverable with additional (corrective) effort |
| atm.gross_cost_eur_per_t_co2 | 150 | assumption | implementation cost of ATM measures per tonne CO2 avoided, before fuel savings; varied 50-500 |
| saf.mandate_steps | 2025: 0.02; 2030: 0.06; 2035: 0.2; 2040: 0.34; 2045: 0.42; 2050: 0.7 | sourced | eu2023 Annex I (minimum shares per period) |
| saf.synthetic_steps | 2030: 0.012; 2032: 0.02; 2035: 0.05; 2040: 0.1; 2045: 0.15; 2050: 0.35 | sourced | eu2023 Annex I (synthetic sub-mandate; 2030-2031 and 2032-2034 as period averages) |
| saf.base_share_2024 | 0.006 | sourced | easa2025refuel (SAF about 0.6 % of EU aviation fuel in 2024) |
| saf.theta_bio | 0.75 | assumption | life-cycle reduction of bio-SAF (HEFA from wastes about 80 %, RED minimum 65 %); varied 0.65-0.85 |
| saf.theta_ptl | 0.85 | assumption | life-cycle reduction of e-SAF (RFNBO minimum 70 %); varied 0.70-0.95 |
| saf.price_fossil_eur_t | 734 | sourced | easa2025prices (2024 reference price, conventional aviation fuel) |
| saf.price_bio_eur_t | 2085 | sourced | easa2025prices (2024 reference price, aviation biofuels) |
| saf.price_ptl_eur_t | 7695 | sourced | easa2025prices (2024 reference price, synthetic aviation fuels) |
| saf.disruption_r_range | 0.57, 0.95 | derived | martulli2025 (EU shortfall 18-43 % in 2030 and 5-35 % in 2050) |
| saf.alt_supply_cap_share | 0.05 | assumption | additional SAF obtainable in a year (imports, book-and-claim) as share of that year's fuel demand |
| zep.entry_year | 2035 | assumption | earliest plausible EIS of battery/hybrid regional aircraft; Airbus postponed ZEROe in 2025 |
| zep.penetration_2050 | 0.4, 0.08, 0.0, 0.0 | assumption | range-feasibility ceilings by band (akinola2025, uddin2025); varied in sensitivity |
| zep.upstream_factor | 2035: 0.3; 2050: 0.1 | assumption | well-to-wake intensity of the ZEP energy carrier relative to fossil kerosene for the same flight |
| corrective.advance_years | 6 | assumption | years by which a retirement is advanced |
| corrective.slots_share_of_fleet | 0.03 | assumption | extra deliveries obtainable per year as share of the band's fleet (manufacturer backlogs); varied 0.01-0.06 |
| corrective.discount_rate | 0.07 | assumption | airline weighted average cost of capital |

| Parameter | Value | Status | Source or rationale |
|---|---|---|---|
| corrective.amortisation_years | 20 | assumption | economic life used in the capital recovery factor |
| corrective.penalty_eur_per_t | 1.0e5 | assumption | penalty on uncompensated emissions; only flags infeasibility |

## References


Akinola, A.C., Langner, J.E.B., Hecker, P., Feuerle, T., Degenhardt, I., 2025. Conceptual feasibility study of integration of aircraft with new propulsion technologies in current airline flight schedules. Aerospace Research Communications 3, 14713.

Bergero, C., Gosnell, G., Gielen, D., Kang, S., Bazilian, M., Davis, S.J., 2023. Pathways to net-zero emissions from aviation. Nature Sustainability 6, 404–414. doi:10.1038/s41893-022-01046-9.

Costa-Alves, I., Gourdain, N., Gallard, F., Gazaix, A., Kambiri, Y.A., Druot, T., 2026. Numerical optimization of aviation decarbonization scenarios: balancing traffic and emissions with maturing energy carriers and aircraft technology. Applied Energy URL: https://www.sciencedirect.com/science/article/pii/S0306261926002837. in press.

Dray, L., Schäfer, A.W., Grobler, C., Falter, C., Allroggen, F., Stettler, M.E.J., Barrett, S.R.H., 2022. Cost and emissions pathways towards net-zero climate impacts in aviation. Nature Climate Change 12, 956–962. doi:10.1038/s41558-022-01485-4.

EASA, 2025a. 2024 Aviation Fuels Reference Prices for ReFuelEU Aviation. Technical Report. European Union Aviation Safety Agency. Cologne.

EASA, 2025b. ReFuelEU Aviation Annual Technical Report 2025. Technical Report. European Union Aviation Safety Agency. Cologne.

EUROCONTROL, 2021. $CO_2$ emissions by flight distance. Data Snapshot 4. EUROCONTROL. URL: https://www.eurocontrol.int/publication/eurocontrol-data-snapshot-co2-emissions-flight-distance.

EUROCONTROL, 2024. EUROCONTROL Aviation Long-Term Outlook: Flights and $CO_2$ Emissions Forecast 2024–2050. Technical Report. EUROCONTROL. Brussels. URL: https://www.eurocontrol.int/publication/eurocontrol-aviation-long-term-outlook.

European Union, 2023. Regulation (EU) 2023/2405 on ensuring a level playing field for sustainable air transport (ReFuelEU Aviation). Official Journal of the European Union, L 2023/2405.

ICAO, 2024. CORSIA Default Life Cycle Emissions Values for CORSIA Eligible Fuels. Technical Report. International Civil Aviation Organization. Montréal. CHECK edition year.

ifeu, 2026. Harmonized GHG Emission Factors for Air Travel in Germany. ifeu paper 01/2026. ifeu – Institut für Energie- und Umweltforschung Heidelberg.

Lee, D.S., Fahey, D.W., Skowron, A., Allen, M.R., Burkhardt, U., et al., 2021. The contribution of global aviation to anthropogenic climate forcing for 2000 to 2018. Atmospheric Environment 244, 117834. doi:10.1016/j.atmosenv.2020.117834.

Martulli, A., Brandt, K., Allroggen, F., Malina, R., et al., 2025. The potential scale-up of sustainable aviation fuels production capacity to meet global and EU policy targets. Nature Communications 16, 11619. doi:10.1038/s41467-025-66686-9.

Performance Review Commission, 2021. The Role of ANS in a Decarbonising World. Technical Report. EUROCONTROL. Brussels. URL: https://ansperformance.eu/library/.

Performance Review Commission, 2025. Performance Review Report: An Assessment of Air Traffic Management in Europe During the Calendar Year 2024. Technical Report. EUROCONTROL. Brussels.

Salgas, A., Lafforgue, G., Delbecq, S., Planès, T., 2025. Enhanced marginal abatement cost curves for analysing and designing aviation decarbonisation scenarios. Transportation Research Part D: Transport and Environment URL: https://www.sciencedirect.com/science/article/pii/S1361920925002469.

Schäfer, A.W., Evans, A.D., Reynolds, T.G., Dray, L., 2016. Costs of mitigating $CO_2$ emissions from passenger aircraft. Nature Climate Change 6, 412–417. doi:10.1038/nclimate2865.

Uddin, M.N., Wang, F., 2025. Sustainable aviation fuels: A review of current techno-economic viability and life cycle impacts. Energies 18, 5510.

Voigt, C., Kleine, J., Sauer, D., Moore, R.H., Bräuer, T., et al., 2021. Cleaner burning aviation fuels can reduce contrail cloudiness. Communications Earth & Environment 2, 114. doi:10.1038/s43247-021-00174-y.